

● EASY

● GEOMETRY AND TRIGONOMETRY

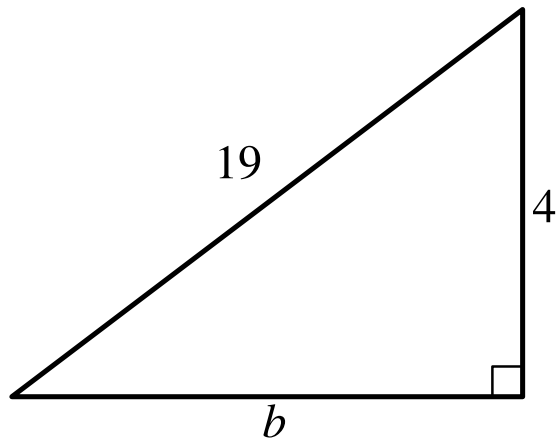
Geometry and Trigonometry

02

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- One angle is a right angle.
- The length of the side opposite the right angle is 19.
- The lengths of the other 2 sides are as follows:
 - b
 - 4
- A note indicates the figure is not drawn to scale.

Which equation shows the relationship between the side lengths of the given triangle?

- A. $4b = 19$
- B. $4 + b = 19$
- C. $4^2 + b^2 = 19^2$
- D. $4^2 - b^2 = 19^2$

$$x - 6^2 + y - 3^2 = 81$$

The graph of the given equation in the xy -plane is a circle. What is the length of the radius of this circle?

- A. 3
- B. 6
- C. 9
- D. 81

A rectangle has a length of 3 units and a width of 39 units. Which expression gives the area, in square units, of this rectangle?

A. $23 + 39$

B. $23 \cdot 39$

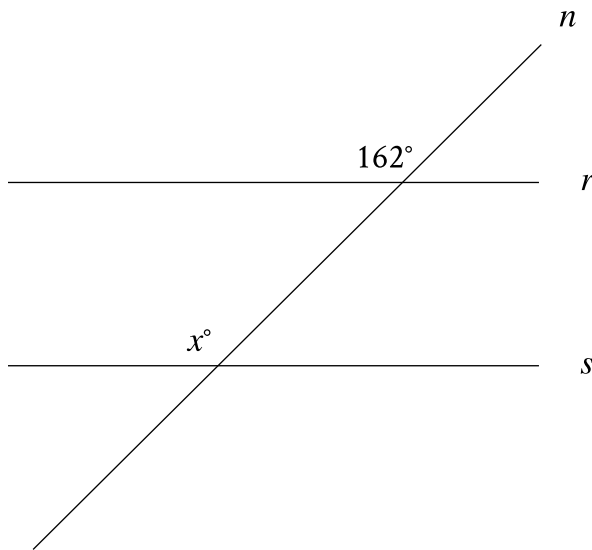
C. $3 + 39$

D. $3 \cdot 39$

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled n , r , and s .
- Line n intersects both line r and line s .
- At the intersection of line n and line r , 1 angle is labeled clockwise from top left as follows:
 - Top left: 162°
- At the intersection of line n and line s , 1 angle is labeled clockwise from top left as follows:
 - Top left: x°
- A note indicates the figure is not drawn to scale.

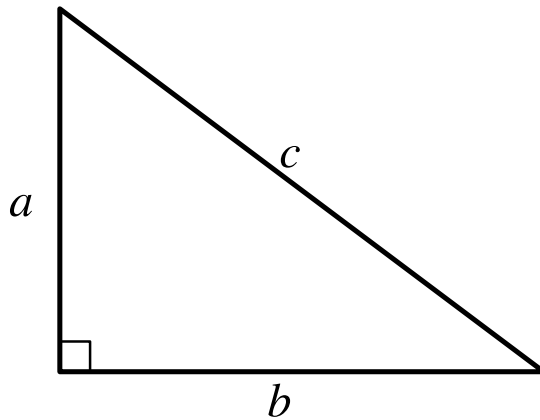
In the figure, line n intersects lines r and s . Line r is parallel to line s . What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0

$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$
$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$
$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- The side lengths are labeled as follows:
 - a
 - b
 - c
- The angle opposite the side labeled c is a right angle.
- A note indicates the figure is not drawn to scale.

For the right triangle shown, $a = 4$ and $b = 5$. Which expression represents the value of c ?

- A. $4 + 5$
- B. $\sqrt{45}$
- C. $\sqrt{4 + 5}$
- D. $\sqrt{4^2 + 5^2}$

Q6

GRID-IN

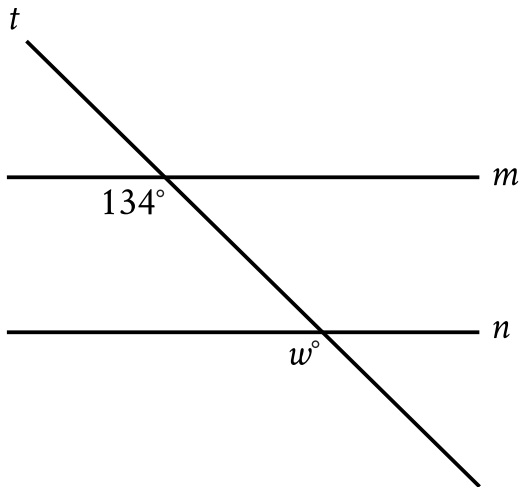
GEOMETRY AND TRIGONOMETRY

A rectangle has a length of 64 inches and a width of 32 inches. What is the area, in square inches, of the rectangle?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

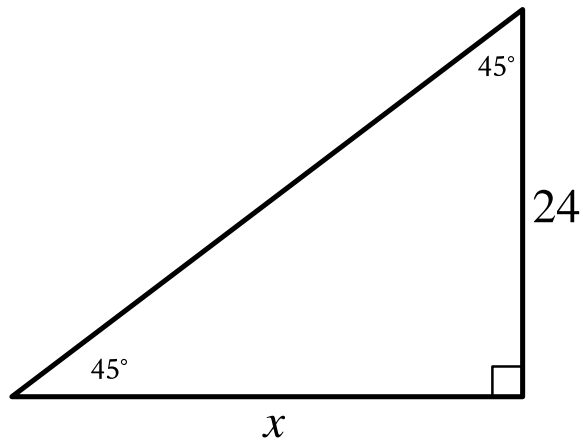


Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t , m , and n .
- Line t intersects both line m and line n .
- At the intersection of line t and line m , 1 angle is labeled clockwise from top left as follows:
 - Bottom left: 134°
- At the intersection of line t and line n , 1 angle is labeled clockwise from top left as follows:
 - Bottom left: w°
- A note indicates the figure is not drawn to scale.

In the figure, line m is parallel to line n . What is the value of w ?

- A. 13
- B. 34
- C. 66
- D. 134



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of the other 2 angles are as follows:
 - 45°
 - 45°
- The lengths of the 2 sides adjacent to the right angle are as follows:
 - x
 - 24
- Note: Figure not drawn to scale.

In the triangle shown, what is the value of x ?

- A. 24
- B. 45
- C. 48
- D. 69

What is the area of a rectangle with a length of 4 centimeters cm and a width of 2 cm?

- A. 6 cm^2
- B. 8 cm^2
- C. 12 cm^2
- D. 36 cm^2

In triangle ABC , the measure of angle B is 52° and the measure of angle C is 17° . What is the measure of angle A ?

- A. 21°
- B. 35°
- C. 69°
- D. 111°